CBSE Class 12 Physics — Atoms

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. When alpha particles are sent through a thin gold foil, most of them go straight through the foil. This is primarily because: (a) alpha particles are positively charged. (b) the mass of an alpha particle is more than the mass of an electron. (c) most of the part of an atom is empty space. (d) alpha particles move with high velocity. [PYQ 2023] Very High

Q2. An electron with angular momentum L moving around the nucleus has a magnetic dipole moment given by: (a) $\frac{eL}{2m}$ (b) $\frac{eL}{3m}$ (c) $\frac{eL}{4m}$ (d) $\frac{eL}{m}$ [PYQ 2023] Very High

Q3. The energy of an electron in the n^{th} orbit of a hydrogen atom is $E_n = -\frac{13.6}{n^2}\ \text{eV}$. The negative sign of the energy strictly indicates that: (a) the electron is free to move. (b) the electron is bound to the nucleus. (c) the kinetic energy of the electron is equal to the potential energy. (d) the atom is radiating energy. [PYQ 2023] Very High

Q4. Which of the following transitions in the Bohr model of a hydrogen atom gives rise to a spectral line of the highest wavelength? (a) $n = 3$ to $n = 1$ (b) $n = 3$ to $n = 2$ (c) $n = 4$ to $n = 1$ (d) $n = 4$ to $n = 3$ [PYQ 2023 / Practice] High

Q5. The radius of the innermost electron orbit of a hydrogen atom is $5.3 \times 10^{-11}\ \text{m}$. What is the exact radius of the $n = 3$ orbit? (a) $1.01 \times 10^{-10}\ \text{m}$ (b) $1.59 \times 10^{-10}\ \text{m}$ (c) $2.12 \times 10^{-10}\ \text{m}$ (d) $4.77 \times 10^{-10}\ \text{m}$ [PYQ 2022] Very High

Q6. Which of the following statements is NOT correct according to the Rutherford model of the atom? (a) Most of the space inside an atom is empty. (b) The electrons revolve around the nucleus under the influence of Coulomb force acting on them. (c) Most part of the mass of the atom and its positive charge are concentrated at its centre. (d) The absolute stability of the atom was established by this model. [PYQ 2020] Very High

Q7. A photon beam of energy $12.1\ \text{eV}$ is incident on a hydrogen atom in its ground state. To which specific orbit will the electron of the H-atom be excited? (a) 2nd (b) 3rd (c) 4th (d) 5th [PYQ 2020] High

Q8. If an electron in a hydrogen atom undergoes a transition from the 1st to the 3rd energy level, by what exact factor does its time period of revolution undergo a change? (a) Remains the same (b) Becomes 3 times the initial value (c) Becomes 9 times the initial value (d) Becomes 27 times the initial value [PYQ 2025-26 SQP] Very High

Q9. A hydrogen atom at its ground state is excited by incident photons of energy $12.75\ \text{eV}$. What is the expected maximum count of distinct spectral lines emitted by this hydrogen atom upon de-excitation? (a) 3 (b) 5 (c) 6 (d) 8 [PYQ 2025-26 SQP] High

Q10. An electron in the Bohr model atom is in the n^{th} orbit. Which of the following statements does NOT comply with the postulates of Bohr's model? (a) The radius of the orbit in terms of de Broglie wavelength is $r = n\lambda/2\pi$. (b) In going from a lower to a higher orbit, both PE and TE increase but KE decreases. (c) If $\text{KE} = 3.4\ \text{eV}$, the corresponding potential energy is $6.8\ \text{eV}$ and Total energy is $10.2\ \text{eV}$. (d) Angular momentum of the electron is quantized. [PYQ 2025-26 SQP] High

Q11. According to Bohr's atomic model, the circumference of the electron's stable orbit is always an integral multiple of what specific physical quantity? [PYQ 2020] Very High

Q12. What is the exact value of the angular momentum of an electron in the second stable orbit ($n = 2$) of Bohr's model of the Hydrogen atom? [PYQ 2021] High

Q13. Define 'Ionisation Energy' of an atom. Write its precise value for a hydrogen atom in its ground state. [PYQ 2023] Very High

Q14. Name the specific spectral series for a hydrogen atom which lies completely in the visible region of the electromagnetic spectrum. [PYQ 2022] Very High

Q15. What is meant by the 'distance of closest approach' when an alpha particle is directed towards a heavy nucleus? [PYQ 2017] High

Q16. Find the maximum possible number of spectral lines emitted when an electron jumps from the fourth orbit ($n = 4$) to the first orbit ($n = 1$) in a hydrogen atom. [PYQ 2016] High

Q17. State Bohr's quantization condition for defining stationary orbits in a hydrogen atom mathematically. [PYQ 2019] Very High

Q18. Assertion (A): The total energy of an electron in a hydrogen atom is strictly negative.
Reason (R): A negative total energy indicates that the electron is bound to the nucleus and requires external energy to be freed. [Practice] Very High

Q19. Assertion (A): The Balmer series of the hydrogen spectrum lies in the visible region.
Reason (R): The transitions for the Balmer series inherently end at the principal quantum number $n = 1$. [Practice] High

Q20. An electron is orbiting in the ground state of a hydrogen atom. What is the fundamental relation between its kinetic energy (K) and potential energy (U)? [Practice] Medium

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2-Mark Questions

Q21. The total energy of an electron in the first excited state of the hydrogen atom is $-3.4\ \text{eV}$. Determine the exact (i) kinetic energy, and (ii) potential energy of the electron in this state. [PYQ 2023 SQP] Very High

Q22. The short wavelength limit for the Lyman series of the hydrogen emission spectrum is $913.4 \backslash AA$. Calculate the exact short wavelength limit for the Balmer series of the hydrogen spectrum. [PYQ 2022 SQP] High

Q23. An α-particle of kinetic energy K is bombarded on a thin gold foil. The distance of the closest approach is r. What will be the new distance of closest approach if an α-particle of double the kinetic energy ($2K$) is bombarded? [PYQ 2017] Very High

Q24. Find the explicit ratio of the longest wavelength in the Lyman series to the shortest wavelength in the Balmer series of the hydrogen spectrum. [PYQ 2022] High

Q25. Find out the exact de Broglie wavelength of the electron orbiting in the ground state of a hydrogen atom. (Given: Radius of ground state $r_1 = 0.53 \backslash\mathrm{AA}$). [PYQ 2017] High

Q26. The electron in a hydrogen atom is typically found at a distance of about $5.3 \times 10^{-11}\ \mathrm{m}$ from the nucleus, which has a diameter of about $1.0 \times 10^{-15}\ \mathrm{m}$. Assuming the hydrogen atom to be a sphere of radius $5.3 \times 10^{-11}\ \mathrm{m}$, what approximate fraction of its volume is actively occupied by the nucleus? [PYQ 2018] Medium

Q27. The ground state energy of a hydrogen atom is $-13.6\ \mathrm{eV}$. If an electron makes a transition from an energy level of $-1.51\ \mathrm{eV}$ to $-3.4\ \mathrm{eV}$, identify the transition levels (n_i and n_f) and name the specific series of the hydrogen spectrum to which it belongs. [PYQ 2023] Very High

Q28. Calculate the de-Broglie wavelength of the electron orbiting in the $n = 2$ state of a hydrogen atom. [PYQ 2016] High

Q29. A hydrogen atom initially in the ground state absorbs a photon which excites it strictly to the $n = 4$ level. Estimate the frequency of the absorbed photon. [PYQ 2018] High

Q30. Write two critical shortcomings of the Rutherford atomic model. Explain briefly how these were effectively overcome by the postulates of Bohr's atomic model. [PYQ 2020, 2015] Very High

Q31. Draw a representative graph showing the variation of the number of alpha particles scattered (N) with the scattering angle (θ) in the Geiger-Marsden experiment. Why are only a very small fraction of the particles scattered at angles $\theta > 90^\circ$? [PYQ 2022] High

Q32. Find the maximum ratio of the maximum to the minimum wavelengths of the Balmer series in the hydrogen spectrum. [PYQ 2022] Medium

3-Mark Questions

Q33. State Bohr's postulate to define stable orbits in a hydrogen atom. How does the de-Broglie hypothesis theoretically explain the absolute stability of these orbits? [PYQ 2018, 2016] Very High

Q34. Using Bohr's postulates, derive the mathematical expression for the frequency of radiation emitted when a hydrogen atom de-excites from energy level n to level $(n-1)$. [PYQ 2022 SQP] High

Q35. The figure below shows a generic energy level diagram. Obtain the mathematical relation between the three wavelengths $\lambda_1, \lambda_2,$ and λ_3 corresponding to the depicted transitions. (Assume a transition from $n_3 \to n_1$ gives λ_3, $n_3 \to n_2$ gives λ_1, and $n_2 \to n_1$ gives λ_2). [PYQ 2016] High

Q36. Using the classical Rutherford model of the atom, derive the explicit expression for the total energy of an electron revolving in a stable circular orbit around the nucleus. [PYQ 2014] Very High

Q37. In the Geiger-Marsden α-particle scattering experiment, what does the 'impact parameter' (b) physically represent? What will be the exact value of b for an alpha particle scattering at (i) $\theta = 0°$, and (ii) $\theta = 180°$? [PYQ 2019] High

Q38. Show mathematically that the radius of the stable orbit in a hydrogen atom varies as n^2, where n is the principal quantum number of the atom, using Bohr's quantization rules. [PYQ 2015, 2014] Very High

Q39. A $12.5\ \mathrm{eV}$ electron beam is used to excite a gaseous hydrogen atom at room temperature. Determine the specific principal quantum number of the excited state and list all the corresponding series of the spectral lines emitted. [PYQ 2017] High

Q40. Calculate the shortest possible wavelength in the Balmer series of the hydrogen atom. In which specific region of the electromagnetic spectrum does this wavelength lie? [PYQ 2015] Medium

Q41. A $12.9\ \mathrm{eV}$ beam of electrons is actively used to bombard gaseous hydrogen at room temperature. Up to which precise energy level would the hydrogen atoms be excited? Calculate the wavelength of the first member of the Paschen series emitted during de-excitation. [PYQ 2014] High

Q42. The kinetic energy of the electron orbiting in the first excited state of a hydrogen atom is exactly $3.4\ \mathrm{eV}$. Determine the de-Broglie wavelength inherently associated with it. [PYQ 2015] High

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5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Describe the Geiger-Marsden α-particle scattering experiment. Draw a schematic plot of the α-particle scattering by a thin foil of gold to show the variation of the number of scattered particles with the scattering angle. (B) Explain briefly how the rare large-angle scattering provides evidence for the existence of the nucleus inside the atom. (C) Explain with the help of the impact parameter picture, how Rutherford scattering serves as a powerful way to determine an upper limit on the absolute size of the nucleus. [PYQ 2019, 2015] Very High

Q44. (A) State Bohr's three postulates for the hydrogen atom model. (B) Using these postulates, systematically derive the expression for the total energy of the electron in the n^{th} stationary state of the hydrogen atom. (C) Explain the physical significance of the total negative energy possessed by the electron. [PYQ 2014] Very High

Q45. (A) Using Bohr's postulates, derive the expression for the orbital period (T) of the electron moving in the n^{th} orbit of a hydrogen atom. (B) Show mathematically that the orbital period varies directly with n^3. (C) Calculate the exact orbital period of the electron in the first excited state ($n = 2$) of a hydrogen atom. [PYQ 2019] High

Q46. (A) Write Rydberg's formula for the wavelengths of the spectral lines of the hydrogen emission spectrum, defining all symbols. (B) Differentiate between the Lyman and Balmer series based on their transitions and spectral regions. (C) Calculate the ratio of the shortest wavelength of the Lyman series to the longest wavelength of the Balmer series. [PYQ 2019 / Practice] High

Q47. A hydrogen atom, initially at rest in its ground state, absorbs a photon of energy 12.75 eV and gets excited. (A) Identify the principal quantum number (n) of the newly excited state. (B) Calculate the total maximum number of spectral lines that can be emitted as the atom de-excites back to the ground state. (C) Identify the transitions corresponding to the shortest and the longest wavelength emitted in this entire process. (D) Calculate the value of the longest wavelength emitted. [PYQ 2025-26 SQP] Very High

Q48. (A) Write two important limitations of the Rutherford nuclear model of the atom. (B) How were these limitations fundamentally overcome by the postulates of Niels Bohr? (C) Obtain Bohr's quantization condition for angular momentum on the basis of the wave nature of an electron as proposed by Louis de Broglie. [PYQ 2020, 2015] Very High

Q49. (A) Define the distance of closest approach (r_0) and derive an expression for it when an alpha particle of mass m and velocity v is bombarded on a nucleus of atomic number Z. (B) An α-particle is accelerated through a potential difference of V volts. Find the expression for its de-Broglie wavelength. (C) Calculate the de-Broglie wavelength of an electron orbiting in the $n = 3$ state of a hydrogen atom. [PYQ 2018 / 2016] High

Q50. Case Study: Energy Levels and Atomic Transitions The energy levels of a hydrogen atom are given by the relation $E_n = -13.6/n^2 \text{ eV}$. When an electron jumps from a higher energy level to a lower energy level, a photon of specific frequency is emitted. Conversely, an atom can absorb a photon of exact energy to jump to a higher level. (a) What is the ionization energy of a hydrogen atom in its first excited state? (b) An electron makes a transition from an energy level of -0.85 eV to -3.4 eV. Calculate the wavelength of the spectral line emitted. (c) Which specific series does the spectral line in part (b) belong to? (d) If the choice of the zero of potential energy is changed (e.g., set to zero at ground state instead of infinity), will the wavelength of the emitted photon in part (b) change? Give a reason. [Practice / PYQ 2023] Very High

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Answer Key

Q1: (c) most of the part of an atom is empty space. **Q2:** (a) $\frac{eL}{2m}$ **Q3:** (b) the electron is bound to the nucleus. **Q4:** (d) $n = 4$ to $n = 3$. (Smallest energy difference corresponds to the highest wavelength, which occurs between adjacent higher orbits). **Q5:** (d) $4.77 \times 10^{-10} \text{ m}$. (

$r_n = n^2 r_1 = 3^2 \times 5.3 \times 10^{-11} = 9 \times 5.3 \times 10^{-11} = 47.7 \times 10^{-11}$ m). **Q6:** (d) The absolute stability of the atom was established by this model. (Rutherford's model could not explain atomic stability). **Q7:** (b) 3rd. ($E_n = E_1 + \Delta E = -13.6 + 12.1 = -1.5$ eV. Since $E_n = -13.6/n^2$, $-1.5 = -13.6/n^2 \Rightarrow n^2 \approx 9 \Rightarrow n = 3$). **Q8:** (d) Becomes 27 times the initial value. ($T \propto n^3$. So $T_3/T_1 = 3^3/1^3 = 27$). **Q9:** (c) 6. ($E_n = -13.6 + 12.75 = -0.85$ eV. This is $n = 4$. Spectral lines = $n(n-1)/2 = 4(3)/2 = 6$). **Q10:** (c) If $\mathrm{KE} = 3.4$ eV, the corresponding potential energy is 6.8 eV and Total energy is 10.2 eV. (Correct values: $\mathrm{PE} = -6.8$ eV, $\mathrm{TE} = -3.4$ eV). **Q11:** De Broglie wavelength (λ). ($2\pi r = n\lambda$). **Q12:** $L = \frac{nh}{2\pi} = \frac{2h}{2\pi} = \frac{h}{\pi}$. **Q13:** Minimum energy required to completely remove an electron from the atom. For H-atom in ground state, Ionisation Energy = $+13.6$ eV. **Q14:** Balmer series. **Q15:** It is the minimum distance from the nucleus to which an α-particle can approach before its entire kinetic energy is converted into electrostatic potential energy and it turns back. **Q16:** Number of lines = $\frac{n(n-1)}{2} = \frac{4(3)}{2} = 6$ lines. **Q17:** $L = mvr = \frac{nh}{2\pi}$, where $n = 1, 2, 3..$. **Q18:** (a) Both A and R are true and R is the correct explanation of A. **Q19:** (c) A is true but R is false. (Balmer series transitions end at $n = 2$, not $n = 1$). **Q20:** $U = -2K$ (Potential energy is negatively twice the kinetic energy).

Q21: Total Energy $E = -3.4$ eV. (i) Kinetic energy $K = -E = +3.4$ eV. (ii) Potential energy $U = 2E = -6.8$ eV. **Q22:** For Lyman limit ($n_1 = 1, n_2 = \infty$): $1/\lambda_L = R(1/1^2) = R$. So $\lambda_L = 1/R = 913.4$ \AA. For Balmer limit ($n_1 = 2, n_2 = \infty$): $1/\lambda_B = R(1/2^2) = R/4$. So $\lambda_B = 4/R = 4 \times 913.4 = 3653.6$ \AA. **Q23:** $K = \frac{1}{4\pi\epsilon_0}\frac{(2e)(Ze)}{r_0}$. Thus $r_0 \propto 1/K$. If kinetic energy is doubled, the new distance of closest approach becomes halved ($r/2$). **Q24:** Longest Lyman ($2 \to 1$): $1/\lambda_L = R(1/1 - 1/4) = 3R/4 \Rightarrow \lambda_L = 4/3R$. Shortest Balmer ($\infty \to 2$): $1/\lambda_B = R(1/4 - 0) = R/4 \Rightarrow \lambda_B = 4/R$. Ratio $\lambda_L/\lambda_B = (4/3R)/(4/R) = 1/3$. **Q25:** Using Bohr's quantization: $2\pi r = n\lambda$. For ground state $n = 1$, $\lambda = 2\pi r_1 = 2 \times 3.14 \times 0.53$ \AA $= 3.33$ \AA. **Q26:** Volume of sphere $\propto r^3$. Fraction $= \frac{V_{nuc}}{V_{atom}} = \frac{R_{nuc}^3}{R_{atom}^3} = \frac{(0.5\times10^{-15})^3}{(5.3\times10^{-11})^3} \approx 10^{-15}$ (or very roughly 10^{-14} to 10^{-15}). **Q27:** -1.51 eV corresponds to $n_i = 3$. -3.4 eV corresponds to $n_f = 2$. Transition is from $n = 3$ to $n = 2$, which belongs to the Balmer series. **Q28:** $v_2 = v_1/2$. $r_2 = 4r_1$. $\lambda = h/mv = 2\pi r_2/2 = \pi r_2 = \pi(4 \times 0.53)$ \AA ≈ 6.66 \AA. **Q29:** $\Delta E = E_4 - E_1 = -0.85 - (-13.6) = 12.75$ eV $= 12.75 \times 1.6 \times 10^{-19}$ J. Frequency $\nu = \Delta E/h = \frac{12.75\times1.6\times10^{-19}}{6.63\times10^{-34}} \approx 3.08 \times 10^{15}$ Hz. **Q30:** (1) Could not explain atomic stability (accelerating electron must radiate energy and spiral into nucleus). (2) Could not explain discrete spectral lines. Bohr overcame this by proposing "stationary orbits" where electrons do not radiate, and emissions only occur during transitions between these discrete orbits. **Q31:** Graph: Y-axis is $N(\theta)$, X-axis is θ. Curve drops sharply from 0° to 180°. Very few scatter $> 90^\circ$ because the entire positive charge and mass is concentrated in a tiny central volume (nucleus); only an alpha particle hitting almost head-on experiences enough repulsive Coulomb force to rebound. **Q32:** Max wavelength ($3 \to 2$): $1/\lambda_{max} = R(1/4 - 1/9) = 5R/36 \Rightarrow \lambda_{max} = 36/5R$. Min wavelength ($\infty \to 2$): $\lambda_{min} = 4/R$. Ratio $= \frac{36/5R}{4/R} = 36/20 = 9/5 = 1.8$.

Q33: Postulate: An electron revolves only in those orbits where its angular momentum is an integral multiple of $h/2\pi$ ($mvr = nh/2\pi$). De-Broglie: Electron is a matter wave. For a stable orbit, the circumference must fit an integral number of wavelengths: $2\pi r = n\lambda$. Since $\lambda = h/mv$, $2\pi r = n(h/mv) \Rightarrow mvr = nh/2\pi$. **Q34:** $\nu = \frac{E_n - E_{n-1}}{h} = \frac{me^4}{8\epsilon_0^2 h^3}\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right]$. For large n, $\frac{1}{(n-1)^2} - \frac{1}{n^2} \approx \frac{2n}{n^4} = \frac{2}{n^3}$. Thus $\nu \approx \frac{me^4}{4\epsilon_0^2 h^3 n^3}$, which exactly matches the classical frequency of revolution $v/2\pi r$. **Q35:** Energy gaps relate as $\Delta E_3 = \Delta E_1 + \Delta E_2$. Since $E = hc/\lambda$, we get $\frac{hc}{\lambda_3} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2} \Rightarrow \frac{1}{\lambda_3} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$. **Q36:** Centripetal force = Electrostatic force: $\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}$. Kinetic energy $K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\epsilon_0 r}$. Potential energy $U = -\frac{e^2}{4\pi\epsilon_0 r}$. Total Energy $E = K + U = -\frac{e^2}{8\pi\epsilon_0 r}$. **Q37:** Impact parameter b is the perpendicular distance of the initial velocity vector of the α-particle from the central axis of the target nucleus. For $\theta = 0^\circ$ (straight through), b is very large (approaching infinity). For $\theta = 180^\circ$ (head-on rebound), $b = 0$. **Q38:** From Bohr's condition: $mvr = \frac{nh}{2\pi} \Rightarrow v = \frac{nh}{2\pi mr}$. Also $\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$. Substitute v: $m\left(\frac{nh}{2\pi mr}\right)^2\frac{1}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$. Solving for r: $r = \frac{\epsilon_0 h^2 n^2}{\pi me^2}$. Since all other terms are constants, $r \propto n^2$. **Q39:** Incident energy = 12.5 eV. Max excitation energy is $-13.6 + 12.5 = -1.1\ \mathrm{eV}$. This is just below $-0.85\ \mathrm{eV}$ ($n = 4$) and above $-1.51\ \mathrm{eV}$ ($n = 3$). Thus, the atom excites to $n = 3$. Series emitted: Lyman ($3 \to 1, 2 \to 1$) and Balmer ($3 \to 2$). **Q40:** Shortest wavelength $\Rightarrow \infty \to 2$. $1/\lambda = R(1/2^2 - 1/\infty) = R/4 \Rightarrow \lambda = 4/R = 4/(1.097 \times 10^7) \approx 3646\ \backslash\mathrm{AA}$. This lies in the visible region (near ultraviolet edge, but primarily classed as visible). **Q41:** $E_n = -13.6 + 12.9 = -0.7\ \mathrm{eV}$. The level $n = 4$ has $-0.85\ \mathrm{eV}$ and $n = 5$ has $-0.54\ \mathrm{eV}$. So it excites to $n = 4$. First member of Paschen ($4 \to 3$): $1/\lambda = R(1/3^2 - 1/4^2) = R(7/144) \Rightarrow \lambda = 144/(7R) \approx 1875\ \mathrm{nm}$. **Q42:** First excited state $\Rightarrow n = 2$. KE = $3.4\ \mathrm{eV}$. $p = \sqrt{2mK}$. $\lambda = \frac{h}{\sqrt{2mK}} = \frac{6.63\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times3.4\times1.6\times10^{-19}}} \approx 6.6\ \backslash\mathrm{AA}$.

Q43: (A) Describe the firing of α-particles from a radioactive source at a thin gold foil, with detection via a movable ZnS screen. Plot N (y-axis) vs θ (x-axis) showing a sharp exponential drop. (B) Very few particles ($\sim 1\ \mathrm{in}\ 8000$) deflect by $> 90^\circ$. This implies the positive charge isn't spread out (as in Thomson's model) but concentrated in a massive tiny core causing strong repulsion. (C) By finding the distance of closest approach for a head-on collision ($b = 0$), where $K_{initial} = U_{electrostatic}$, we calculate $r_0 \approx 10^{-14}\ \mathrm{m}$, establishing the upper limit for nuclear radius. **Q44:** (A) Postulates: 1. Electrons orbit in stable circular paths without radiating. 2. Angular momentum is quantized ($L = nh/2\pi$). 3. Transitions between orbits emit/absorb a photon ($h\nu = E_i - E_f$). (B) Centripetal force = Electrostatic force gives v^2. KE = $mv^2/2 = e^2/8\pi\epsilon_0 r$. PE = $-e^2/4\pi\epsilon_0 r$. TE = $-e^2/8\pi\epsilon_0 r$. Substitute r from quantization condition to get $E_n = -me^4/8\epsilon_0^2 h^2 n^2$. (C) The negative sign physically implies the electron is bound to the nucleus by attractive forces; work must be done to rip it completely away to infinity (where PE=0). **Q45:** (A) $v_n = \frac{e^2}{2\epsilon_0 hn}$ and $r_n = \frac{\epsilon_0 h^2 n^2}{\pi me^2}$. Period $T = \frac{2\pi r_n}{v_n} = 2\pi\left(\frac{\epsilon_0 h^2 n^2}{\pi me^2}\right)\left(\frac{2\epsilon_0 hn}{e^2}\right) = \frac{4\epsilon_0^2 h^3 n^3}{me^4}$. (B) From the formula, $T = (\mathrm{constant}) \times n^3 \Rightarrow T \propto n^3$. (C) For $n = 1$, $T_1 = 1.52 \times 10^{-16}\ \mathrm{s}$. For $n = 2$,

$T_2 = 2^3 \times T_1 = 8 \times 1.52 \times 10^{-16} \approx 1.22 \times 10^{-15}$ s. **Q46:** (A) $\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$, where R is Rydberg constant ($1.097 \times 10^7 \text{ m}^{-1}$). (B) Lyman: ends at $n_f = 1$, falls in Ultraviolet. Balmer: ends at $n_f = 2$, falls in Visible. (C) Shortest Lyman ($\infty \to 1$): $\lambda_{L(min)} = 1/R$. Longest Balmer ($3 \to 2$): $1/\lambda_{B(max)} = R(1/4 - 1/9) = 5R/36 \Rightarrow \lambda_{B(max)} = 36/5R$. Ratio $= (1/R)/(36/5R) = 5/36$. **Q47:** (A) $E_n = -13.6 + 12.75 = -0.85 \text{ eV}$. Thus $n = 4$. (B) Lines = $4(3)/2 = 6$. (C) Shortest wavelength = largest energy difference = $4 \to 1$. Longest wavelength = smallest energy difference = $4 \to 3$. (D) For $4 \to 3$: $\Delta E = -0.85 - (-1.51) = 0.66 \text{ eV}$. $\lambda = hc/\Delta E = 1242/0.66 \approx 1881 \text{ nm}$ (Paschen series). **Q48:** (A) 1. Failed to explain the stability of the atom. 2. Failed to explain the line emission spectrum of hydrogen. (B) Bohr proposed that electrons orbit in 'stationary' states without radiating energy (explaining stability), and only radiate when jumping between these discrete states (explaining line spectra). (C) De Broglie stated matter has wave nature, $\lambda = h/mv$. For a stable orbit, standing waves must form: $2\pi r = n\lambda$. Substituting $\lambda \Rightarrow 2\pi r = nh/mv \Rightarrow mvr = nh/2\pi$. **Q49:** (A) Distance where α-particle's kinetic energy entirely converts to potential energy. $\frac{1}{2}mv^2 = \frac{1}{4\pi\epsilon_0}\frac{(2e)(Ze)}{r_0} \Rightarrow r_0 = \frac{4Ze^2}{4\pi\epsilon_0 mv^2}$. (B) $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}} = \frac{h}{\sqrt{4meV}}$. (C) $n = 3$, $v_3 = v_1/3 \approx 2.18 \times 10^6/3 \approx 7.26 \times 10^5 \text{ m/s}$. $\lambda = h/mv = \frac{6.63\times10^{-34}}{9.1\times10^{-31}\times7.26\times10^5} \approx 10 \text{\AA}$ (or using $2\pi r_3 = 3\lambda$). **Q50:** (a) First excited state is $n = 2$ with energy -3.4 eV. Ionization energy is $+3.4 \text{ eV}$. (b) -0.85 eV is $n = 4$. -3.4 eV is $n = 2$. $\Delta E = 3.4 - 0.85 = 2.55 \text{ eV}$. $\lambda = 1242/2.55 \approx 487 \text{ nm}$. (c) Since it ends at $n = 2$, it belongs to the Balmer series. (d) No. The wavelength depends strictly on the *difference* in energy (ΔE). Shifting the zero reference shifts both levels equally, keeping ΔE identical.
